

and the access node reduces. This accounts for better transmission quality and performance, reduced delay and better coverage especially at cell edge. It also provides a great business solution as the user gets flexible and quality services at a reduced cost.

In late 2014, the second edition of the annual Small Cells India 2014 event took place in Mumbai, where players like Huawei, Aricent, iBwave and members of Cellular Operators Association of India (COAI) came together to understand how to cost-effectively optimise small cell networks in India.

How is 5G shaping up

The fifth generation of mobile communication is already in its development phase around the world. Any new telecommunication technology including 5G (excluding standalones like Bluetooth, Zigbee or Wi-Fi) will be very similar to LTE, foresees Nagaraja Upadhyaya, vice president of network product line, Huawei Technologies Co. Ltd. Although the industry's estimated year of commercial introduction of 5G is 2020, Huawei has already collaborated with Mega-

When will we see 5G

Historically, successive generations of mobile technology came out every 10 years. 1G was introduced in 1981, followed by the second-generation rollout in 1991. The first 3G system showed its head in 2001 and 4G systems compliant with international mobile telecommunications-Advanced (IMT-Advanced) were standardised in 2012. We can speculate that 5G will be ready for primetime in 2021. Right now, teams worldwide are working on the concept.

4G standardised in 2012; how have we not seen it yet

There are several factors that contributed towards the success of 4G telecommunication technology around the world. Better speed, voice and video clarity, buffer and bus efficiency, battery life and security, speedy adoption of smartphones by customers and replacement of landlines, and reduced latency are some of the major factors that drove this movement. This also positively affected the business sector.

But in India, 3G and 4G have not reached their prime yet. Several rural and some urban areas still rely on 2G and previous technologies for mobile communication. The high cost and complicated procedures for buying the spectrum results in operators having very little time and money to invest on infrastructure development and deployment of new technologies.

Fon, and is targeting demonstration of this technology during FIFA World Cup 2018. 5G would be expensive and more beneficial for the business sector, rather than the end consumer, predicts Moorthy.

Apart from having better speed, this technology is going to focus on having low latency (of a few milliseconds), high quality of data transfer, better bandwidth and network capacity (number of connections that can be implemented). These features are

an asset to business models for cyber physical systems or the Internet of Things (IoT) and machine-to-machine (M2M) communication.

Next Generation Mobile Networks (NGMN) expects customer requirements of 5G in the 2020+ timeframe to result in the accommodation of massive traffic growth and high-density demand, a wide variety and variability of services consumed, new use cases such as machine-type communication (M2M and the IoT) and stringent demands for real-time communications.

NGMN has finalised and plans to announce the publication of its 5G white paper at the NGMN industry conference and exhibition, scheduled for March 24-25, 2015, in Frankfurt, Germany. Board of directors, senior industry and research representatives as well as subject matter experts plan to discuss all related technology and eco-system aspects. The conference might provide a good overview on 5G requirements, its technology evolution, architecture principles, future services, spectrum demand and intellectual property rights policy.

Technological hiccups

Since many mobile phone users in India still use GSM and 2G technology, it will take a while to adopt 4G and 5G; an overlap period of these technologies is expected to be high. This

Who on Earth uses WiMAX

Worldwide interoperability for microwave access, popularly known as WiMAX, is a communication standard that descends from W-Fi technology. WiMAX was created by WiMAX forum in early 2000s, even before the advent of LTE. WiMAX is similar to LTE in that these both claim to belong to the fourth generation and support multiple-input and multiple-output (MIMO) antenna technology and orthogonal frequency division multiplexing (OFDM) modulation technology.

But it failed to take off like the latter due to several reasons that include:

- LTE is a natural technology evolution of previously used mobile technologies. It is compatible to, and provides easy migration path from standards like GSM, GPRS, UMTS, EDGE, WCDMA, HSPA, CDMA-one, CDMA2000, EV-DO, EV-DV and SC-CDMA for operators, and components and device manufacturers. The smooth transition from 3G to 4G also leads to a better user experience. In addition to this, LTE also supports roaming services better than WiMAX.

- LTE is backed by 3rd generation partnership project (3GPP), one of the strongest organisations for mobile telecommunication. "3GPP forum is very well accepted among telecom components and chips manufacturers, handset vendors, test and measurement companies, and infrastructure developers," says Jayanth Ramachandran.

- Performance of LTE is also found to be better than that of WiMAX. It handles speed of about 450kmph, while WiMAX can go only to about 120kmph. The former offers better mobility to customers.

- "Managing networks becomes easier with LTE," says Moorthy. "It provides an end-to-end IP network solution," he adds.

But the cost of deployment for WiMAX networks is much lesser as compared to LTE networks. According to Upadhyaya, "WiMAX is the best choice for private mobile broadband wireless networks where cost is a major concern".